REV	ECO#	DESCRIPTION OF REVISION
01		HRE-FREE VERSION OF 160-04644. NO 3D CHANGES. KHELBERT 06/25/25

NOTES: (UNLESS OTHERWISE SPECIFIED)

1. GEOMETRIC DATA FILES PROVIDED BY APPLE INC. SHALL BE USED FOR MASTER DATA. A DRAWING OF EQUAL FILENAME AND REVISION TO THE GEOMETRIC DATA SHALL BE USED TO DETERMINE GEOMETRIC CHARACTERISTICS AND TOLERANCES. DRAWING DIMENSIONS ARE SUBORDINATE TO GEOMETRIC DATA UNLESS SPECIFIED ON DRAWING TO BE MASTER, AS MAY BE THE CASE OF CABLE LENGTHS OR TABULATED VARIABLE DIMENSIONS. THE SUPPLIER MUST COMPLY WITH THE STANDARDS PERTAINING TO THE TOLERANCE METHOD APPLIED.

2. GD&T - GEOMETRIC TOLERANCE METHODS IN ACCORDANCE TO ASME Y14.5-2009. DATUM REFERENCE FRAMES RANK AS DATUM A PRIMARY, B SECONDARY AND C TERTIARY. FEATURE CONTROL FRAMES MAY RE-ORDER DATUMS PER THE FEATURES DATUM DEPENDENCE. DIMENSIONS THAT LOCATE GEOMETRIC TOLERANCE CHARACTERISTICS ARE BASIC (EXACT). NO TOLERANCES APPLY TO BASIC DIMENSIONS.

3. DIRECT TOLERANCE DIMENSIONING - INTERPRET DIMENSIONS THAT DO NOT LOCATE GD&T CHARACTERISTICS OR GD&T CONTROLLED FEATURES AS THE DIRECT TOLERANCE DIMENSION SYSTEM. THE LOCAL PLUS+/MINUS-, LOCAL LIMIT AND TITLE BLOCK TOLERANCES APPLY.

4. IN THE ABSENCE OF DRAWING DIMENSIONS AND DATUMS, THE GEOMETRIC DATA IS BASIC. THE PLANES OF THE ABSOLUTE COORDINATE SYSTEM DENOTE THE DATUM REFERENCE PLANES AS X-Y = DATUM A, Y-Z = B AND X-Z = C. ALL GEOMETRIC DATA SURFACES ⌓ 0.2 A B C.

5. PERFECT ORIENTATION AND/OR PERFECT LOCATION AT MMC AS REQUIRED FOR THE INTERRELATIONSHIP OF ALL DATUM FEATURES OF SIZE.

6. ON PARTS SUBJECT TO FREE STATE VARIATION, ALL TOLERANCES APPLY WHEN THE PART IS RESTRAINED ACCORDING TO FUNCTIONAL MATING CONDITIONS AS DEFINED BY THE DATUM REFERENCE FEATURE RANK OR FEATURE CONTROL FRAME DATUM RANK PER THE FEATURE INSPECTED. CONSULT PROPER APPLE INC. MANUFACTURING QUALITY ENGINEER (MQE) FOR APPROVAL/CLARIFICATION ON INDIVIDUAL PARTS.

7. DIMENSIONS DESIGNATED WITH THE SYMBOL (SPC) SHALL BE MONITORED WITH STATISTICAL PROCESS CONTROLS. THE PROCESS CONTROL DATA (CP, CPK, ETC.) OF THE PART MUST BE VALIDATED AND APPROVED BY APPLE INC. MQE.

8. DIMENSIONS DESIGNATED WITH THE (FAI) SYMBOL SHALL BE FIRST ARTICLE INSPECTION DIMENSIONS. THREE SAMPLE PARTS MUST BE FULLY INSPECTED. FAI REPORT MUST BE VALIDATED AND APPROVED BY APPLE INC. MQE.

9. ALL TOOLING, FIXTURING AND OTHER UNIQUE ITEMS THAT ARE USED TO CREATE THIS PART ARE THE PROPERTY OF APPLE INC. AND SHALL BE PERMANENTLY MARKED IN ACCORDANCE WITH DOCUMENT SRAS-2024.

10. THE DESIGN OF ALL TOOLING OR FIXTURING REQUIRED FOR THE MANUFACTURING OR VERIFICATION OF THE PART SHALL BE APPROVED BY THE APPROPRIATE APPLE INC. ENGINEER PRIOR TO TOOL OR FIXTURE FABRICATION.

11. PART MUST BE PACKAGED AND LABELED IN A MANNER TO ENSURE THAT EACH INDIVIDUAL PIECE CANNOT BE COMPRESSED, DAMAGED, CRUSHED OR ALTERED FROM ITS ORIGINAL STATE DUE TO SHIPPING OR REMOVAL FROM PACKAGING.

12. PART TO BE FREE FROM CONTAMINANTS, LUBRICANTS, AND FLAKES AS SPECIFIED BY APPLE INC. MDE OR MQE.

13. ALL DIMENSIONS APPLY AFTER FINISHING UNLESS OTHERWISE SPECIFIED.

14. ALL HOMOGENEOUS MATERIALS MUST COMPLY WITH THE APPLE INC. REGULATED SUBSTANCES SPECIFICATION, 069-0135.

    ALL ADHESIVES, COATINGS AND PAINTS, PRINTING INKS AND CLEANING AGENTS USED IN THE MANUFACTURING OF THIS PART MUST COMPLY WITH THE APPLE INC. VOC SPECIFICATION, 099-22549.

    ALL MATERIALS WITH RECYCLED OR RENEWABLE CONTENT MUST COMPLY WITH THE APPLE INC. RECYCLED AND RENEWABLE MATERIAL SPECIFICATION, 099-15583.

15. PART MUST COMPLY WITH APPLE INC. SAFETY SPECIFICATION, 099-39074:
    * EXEMPT COMPONENT

16. ALL SURFACES COSMETIC PER CLASS C, APPLE INC. DOCUMENT 069-1770.

17. PART MUST SURVIVE 24HR SALT SPRAY TEST PER APPLE INC. SPECIFICATION 080-2703.

18. APPROXIMATE PART VOLUME: 92.69 mm^3
    APPROXIMATE PART SURFACE AREA: 137.35 mm^2
    APPROXIMATE PART MASS: 0.70 g

19. MATERIAL: NeFeB, (FAI 16)
    GRADE: N50H-100RC(REE)
    RESIDUAL FLUX DENSITY (Br): 13.9-14.4 kGs
    MAX ENERGY PRODUCT (BH)max: 47-51 MGOe
    Hci: > 16KOe
    SHOULD MAINTAIN ABOVE VALUES AND MAGNETIC MOMENT ±5% AFTER HEAT SOAK 85°C/85% RH, 72 HRS.

20. MAGNETIC MOMENT:
    MMz: 12.40 ±7.5% μWb*cm (FAI 17) (SPC 0)

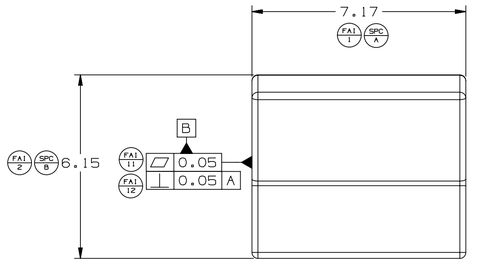
21. PLATING: SILVER NICKEL
    THICKNESS REQUIREMENT:
    LAYER 1, Ni: 5 μm MIN, 10 μm MAX (FAI 18)
    LAYER 2, Cu: 5 μm MIN, 10 μm MAX
    LAYER 3, Ni: 5 μm MIN, 10 μm MAX
    TOTAL: 15 μm MIN, 30 μm MAX
    COLOR: SILVER
    PLATING THICKNESS MEASURED AT 5 POINTS PER MAGNET, 5 MAGNETS PER BATCH

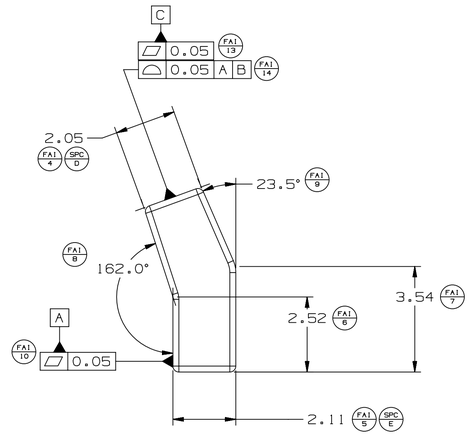
22. PART MAGNETIZED WITH FIELD > 3.5 TESLA.

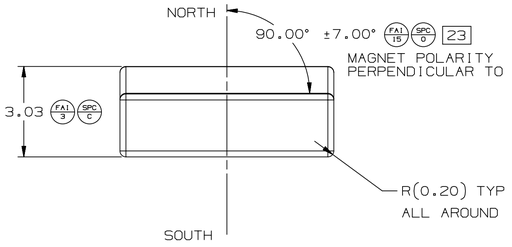
23. MAGNETIZE IN AXIAL DIRECTION WITH INDICATED POLARITY.

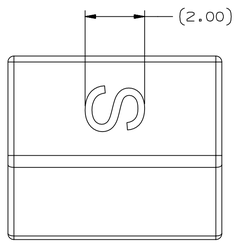
24. MARK SOUTH POLE WITH LASER OR EQUIVALENT PERMANENT MARKING METHOD.

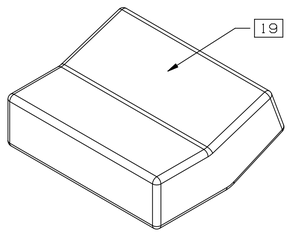
25. FOLLOW APPLE INC. MAGNET ERS 080-02981.

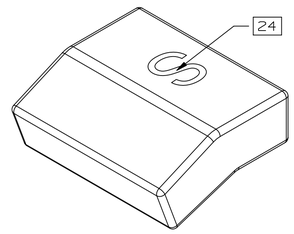












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DIMENSIONS ARE IN MILLIMETERS TOLERANCES X.X ±0.2 X.XX ±0.10 X.XXX ±0.050 ANGLES ±0.5° DO NOT SCALE DRAWINGS		TITLE: MAGNET, REAR, HRE-FREE, B735
		DRAWING NUMBER: 160-06269 REV. 01
THIRD ANGLE PROJECTION	SIZE D	SCALE: NONE SHT 1 OF 1

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